

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA0304

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr DUMPALA PRASANTH

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

V Sunil Kumar

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q1. Reward Trend and Policy Improvement

Answer

In Reinforcement Learning, reward represents the feedback received by an agent after performing an action. The objective of the agent is to maximize its cumulative reward over multiple episodes.

In this case, the reward increases gradually from 10 to 45. This indicates that the agent is improving its decision-making ability through repeated interaction with the environment.

Training Stage	Reward	Interpretation
Initial	10	Limited knowledge
Early training	20	Initial learning
Middle training	30	Better decisions
Later training	40	Improved policy
Final	45	Strong performance

The improvement is:

Therefore, the reward has increased by 35 points.

Initially, the agent may take random or ineffective actions because it has little knowledge of the environment. As training continues, it learns from rewards and penalties and gradually selects better actions.

An increasing reward indicates:

- Better decision-making.
- Improvement in the learned policy.
- More effective action selection.
- Reduction in poor decisions.
- Progress toward convergence.

If the reward becomes stable around a high value after further training, it provides stronger evidence that the policy has converged.

Conclusion: The increase in reward from 10 to 45 demonstrates significant learning and policy improvement. However, increasing reward alone does not prove that the policy is globally optimal.

Q2. Effect of Epsilon on Exploration and Exploitation

Answer

The ϵ -greedy strategy balances exploration and exploitation.

- Exploration: Trying new actions to discover better possibilities.
- Exploitation: Selecting the best-known action.

The probabilities are:

ϵ	Exploration	Exploitation
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0.1	10%	90%
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0.3	30%	70%
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0.5	50%	50%
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When $\epsilon = 0.1$, the agent exploits its learned knowledge 90% of the time. Therefore, it can frequently select actions that have already demonstrated good rewards.

At $\epsilon = 0.3$, exploration increases, which may help discover new actions but can also reduce immediate reward.

At $\epsilon = 0.5$, the agent explores half of the time. This means it frequently selects actions that may not be optimal, resulting in lower cumulative reward.

The observed relationship is:

Conclusion: $\epsilon = 0.1$ provides the best performance in this experiment because it gives greater importance to exploitation while still allowing some exploration.

Q3. Q-Learning and SARSA Performance

Answer

Q-Learning and SARSA are temporal-difference Reinforcement Learning algorithms.

Q-Learning is an off-policy algorithm:

SARSA is an on-policy algorithm:

The observed results show that Q-Learning consistently produces slightly higher rewards at 100, 200, and 300 episodes.

Episodes Q-Learning SARSA

100	Higher	Lower
200	Higher	Lower
300	Higher	Lower

Q-Learning uses the maximum estimated Q-value for the next state, whereas SARSA uses the actual next action selected by its current policy.

The consistently higher reward indicates that Q-Learning performs better under the given conditions.

However, reward values alone cannot completely prove faster convergence. A proper convergence analysis should also consider how quickly each algorithm reaches a stable reward level.

Conclusion: Q-Learning shows better performance than SARSA in the given experiment and appears to learn the better policy more quickly.

Q4. Effect of Discount Factor

Answer

The discount factor determines the importance given to future rewards.

The return is calculated as:

γ Main Focus

0.2 Immediate rewards

0.5 Immediate and future rewards

0.9 Long-term rewards

When $\gamma = 0.2$, future rewards are heavily discounted. Therefore, the agent mainly focuses on immediate benefits.

When $\gamma = 0.5$, the agent gives moderate importance to future outcomes.

When $\gamma = 0.9$, future rewards have much greater influence. The agent can therefore select an action that provides a smaller immediate reward but produces a larger long-term reward.

The observed result is:

Conclusion: $\gamma = 0.9$ is the most suitable value among the tested values because it produces the highest average reward and encourages long-term decision-making.

Q5. Success Rate Improvement

Answer

The success rate increases from 45% to 92%, indicating significant improvement in the agent's performance.

The improvement is:

Therefore, the success rate increases by 47 percentage points.

The relative improvement is:

Initially, the 45% success rate indicates that the agent has limited knowledge and has not yet developed an effective policy.

With additional training, the agent receives more experience and learns which actions produce successful outcomes. By 200 episodes, the success rate reaches 92%.

This indicates:

- Better action selection.
- Improved policy quality.
- Greater consistency.
- Better understanding of the environment.
- Increased task-solving ability.

However, a 92% success rate does not automatically prove that the policy is optimal. To establish optimality, performance should remain stable during additional training and ideally be compared with the theoretical maximum or an optimal baseline.

Conclusion: The increase from 45% to 92% strongly indicates policy improvement and suggests that the agent may be approaching a near-optimal policy.

Q6. Effect of Learning Rate

Answer

The learning rate determines how much the agent changes its existing knowledge based on new experience.

A simplified update can be represented as:

Learning Rate Learning Behavior

0.01 Slow but stable

0.05 Balanced

0.1 Fast but potentially unstable

With a learning rate of 0.01, the updates are very small. This provides stability but may result in slow learning.

At 0.05, the updates are moderate. The agent can learn at a reasonable speed without producing excessive fluctuations.

At 0.1, updates are large. Although learning may initially be faster, the agent can overreact to individual experiences and become unstable.

Since the highest stability is observed at 0.05, it provides the best balance between speed and stability.

Conclusion: A learning rate of 0.05 is the best choice for this experiment because it provides effective learning while maintaining stable convergence.

Q7. Comparison of Exploration Strategies

Answer

The three exploration strategies are ϵ -greedy, Softmax, and random exploration.

ϵ -Greedy

ϵ -greedy selects the best-known action most of the time and randomly explores with probability ϵ .

It provides a simple balance between exploration and exploitation.

Softmax

Softmax assigns probabilities to actions according to their Q-values:

Actions with higher Q-values receive higher probabilities.

Random Exploration

Random exploration selects actions without considering their learned values. Therefore, the agent may frequently choose poor actions.

Strategy	Uses Learned Q-Values	Exploration
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ϵ -Greedy	Yes	Controlled
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Softmax	Yes	Probability-based
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Random	No	Very high
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Since ϵ -greedy obtains the highest cumulative reward, it provides the most effective balance for this particular environment.

Conclusion: ϵ -greedy achieves better performance because it exploits good actions most of the time while still providing enough exploration to discover alternatives.

Q8. Reduction in Maze Navigation Steps

Answer

The agent initially requires 50 steps to reach the goal and later requires only 15 steps.

The reduction is:

The percentage reduction is:

Therefore, the agent improves its navigation efficiency by 70%.

Initially, the agent does not know the best route and explores different paths. Some of these paths may be longer or lead away from the goal.

As training continues, the agent learns which states and actions lead toward the destination. It gradually avoids unnecessary movements and selects shorter paths.

The reduction in steps indicates:

- Improved navigation.
- Better action selection.
- Improved policy.
- Increased efficiency.
- Learning from previous experiences.

However, reaching 15 steps after 300 episodes does not alone prove optimal convergence. If the agent continues to take approximately 15 steps over many additional episodes, it would provide stronger evidence of convergence.

Conclusion: The reduction from 50 to 15 steps represents a 70% improvement and strongly indicates policy improvement and more efficient navigation.

Q9. SARSA and Discount Factor

Answer

The discount factor determines the importance given to future rewards in SARSA.

The SARSA update is:

The tested values are:

γ	Future Reward Importance	Decision Behavior
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0.4	Low	Short-term
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0.7	Medium	Balanced
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0.95	High	Long-term
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With $\gamma = 0.4$, the agent strongly prioritizes immediate rewards.

With $\gamma = 0.7$, the agent considers both immediate and future rewards.

With $\gamma = 0.95$, future rewards have a much greater influence on the current decision.

Since $\gamma = 0.95$ produces the highest long-term reward, the environment appears to benefit from long-term planning.

For example, an action that produces a small immediate reward may lead to a much larger reward later. A high discount factor helps the agent recognize this long-term benefit.

Conclusion: $\gamma = 0.95$ is the most effective value in this experiment because it produces the highest long-term reward and encourages long-term decision-making.

Q10. Reward Stability in a Stochastic Environment

Answer

In a stochastic environment, randomness can cause the reward to fluctuate, particularly during the early stages of training.

A typical training pattern is:

Early episodes → High reward fluctuations

Intermediate episodes → Gradual improvement

Around 500 episodes → Reduced fluctuations

After 500 episodes → Stable rewards

During early training, the agent has limited knowledge and explores different actions.

Randomness in the environment also causes variations in the rewards.

As the agent gains experience, it learns which actions are more effective. Its policy becomes increasingly consistent.

After approximately 500 episodes, the rewards stabilize. This indicates that the agent has developed a more reliable strategy.

Stable rewards can indicate:

- Improved policy.
- Consistent decision-making.
- Reduced unnecessary exploration.
- Better handling of environmental randomness.
- Increased robustness.
- Possible convergence.

However, stable rewards do not automatically mean that the policy is optimal. A policy can converge to a stable but suboptimal solution.

Conclusion: The stabilization of rewards after 500 episodes indicates that the agent has learned a more consistent and robust policy. It provides evidence of convergence and learning stability, although additional comparison with an optimal or baseline policy is needed to prove optimality.